

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: appliance connector

Assistant: To better understand your needs, could you please clarify:

1. What is the exact use of the polymer in your appliance connector? For example, is it for the connector housing, insulation, seals/gaskets, or strain relief?
2. Where will this connector be used in your appliance solution? For instance, is it for internal connections (like PCB or motor), external power input/output, or another specific function?

Your answers will help identify the most suitable material options for your application.

User: connector housing, for refrigerator

Assistant: Thank you for clarifying. To proceed, could you please provide:

1. What is the end-use scenario for this connector housing in the refrigerator? For example, is it used for connecting the compressor, motor, sensors, or PCB modules?
2. What are the typical operating conditions for this connector housing? For instance, is it exposed to moderate heat, vibration, humidity, or located near the compressor/motor?

Additionally, please let me know:

3. What is the minimum wall thickness of your connector housing part?

4. Do you have any preferred base materials for this application (such as specific polymers)?
5. Are you open to alternative material suggestions if there is no strict preference?

These details will help identify the most suitable material options for your connector housing.

User: i can not provide this information and i dont have any preference and open to suggestions

Assistant: Here are the key material requirements (CTQs) assumed for your refrigerator connector housing application:

1. Processing Requirement: Injection Molding
2. Flame retardant: Halogenated flame retardant
3. Flame Rating: UL 94 V-0 at 0.75 mm
4. Relative Thermal Index - Electrical (RTI Elec): $\geq 120^{\circ}\text{C}$
5. Comparative Tracking Index (CTI): $\geq 250\text{ V}$
6. Glow Wire Flammability Index (GWFI): 850°C
7. Glow Wire Ignition Temperature (GWIT): 775°C
8. Minimum Thickness: 0.75 mm
9. High Flow (for thin-wall molding)
10. Dimensional stability
11. Electrical insulation
12. Low humidity absorption

Are all of these requirements correct for your connector housing application? If you need to adjust any, please let me know.

User: looks good recommend the grades

Assistant: Based on your requirements for a refrigerator connector housing, here are four grades that may be suitable options:

1. FRIANYL® A3 GF25 V0XI NC 1101/LA
2. FRIANYL® A3 GF25 V0XI
3. Zytel® FR50 NC010
4. Zytel® FR50 NC010A

All of these grades are halogenated flame-retardant PA66 materials, designed for injection molding, and meet key requirements such as UL 94 V-0 at 0.75 mm, RTI Electrical $\geq 120^{\circ}\text{C}$, GWIT/GWFI, CTI ≥ 250 V, high flow for thin-wall molding, dimensional stability, and electrical insulation. FRIANYL® A3 GF25 V0XI NC 1101/LA is especially noted for its high flow and proven use in appliance connector housings.

These suggestions are based on technical data and typical application requirements. Please review them for suitability in your design, and ensure compliance with all applicable standards and regulations. Final material selection and product design decisions rest with you.

Note: This recommendation is generated by an AI assistant and should be considered purely informational.